TPU communication lowering & routing

Standard XLA HLO and custom kernel paths converge at the low-level optimizer.

1. Model Authoring Layer

JAX / PyTorch:
`jax.lax.all_gather, pmean, torch_xla`

Pallas / Custom Kernel:
`pl.pallas_call(), pltpu.async_copy`

Path 1: Standard XLA HLO Path

StableHLO / JAXPR Graph

XLA HLO Collectives:
`all-gather, all-reduce, all-to-all`

XLA Backend Emitter:
`lowering_emitter.cc / LoweringUtil`

Path 2: Custom Kernel Path (Pallas / LLO)

Mosaic MLIR:
`tpu.enqueue_dma / tpu.wait_dma2`

HLO Wrapper:
`tpu_custom_call`
(Bypasses HLO Collectives)

Mosaic lower_to_llo:
Directly calls LLO Region Builder

2. Low-Level Optimizer (LLO)

`LloRegionBuilder:: EnqueueDmaGeneral()`

VLIW Bundle Packing &
DMA Descriptor State

3. Hardware DMA, Routing & Network

Routing Algorithms:
- Unicast Routing
- Multicast Routing

Populates

DMA Descriptor:
`addrs, sync_flags, dma_type, target_chip_id / multicast_id`

Indexes by ID

On-Chip Routing Tables:
Egress & Link Next-Hop

Physical BoardFly Interconnect:
VC0 → VC1 Switch → Dest Memory

Lowering / execution flow

Routing-table population